# Unconventional prime-field ECDLP leads: scouting round, 2026-11-01

**Status.** Scouting note. Nothing here is a measurement, a proof, or a change of research state. Every heuristic is numbered and unproven. Every proposed disposition is a proposal for the Coordinator.

**Question.** Which mechanisms for prime-field ECDLP, drawn from fields outside elliptic-curve cryptography, are absent from the lead atlas, and what first step would decide each one?

**Outcome.** Three new lead rows, two proposed amendments to existing rows, three quick kills recorded without rows, and nine candidates that collided with existing leads. None of the new rows is predicted to beat matched rho. Each one carries a discriminating first step.

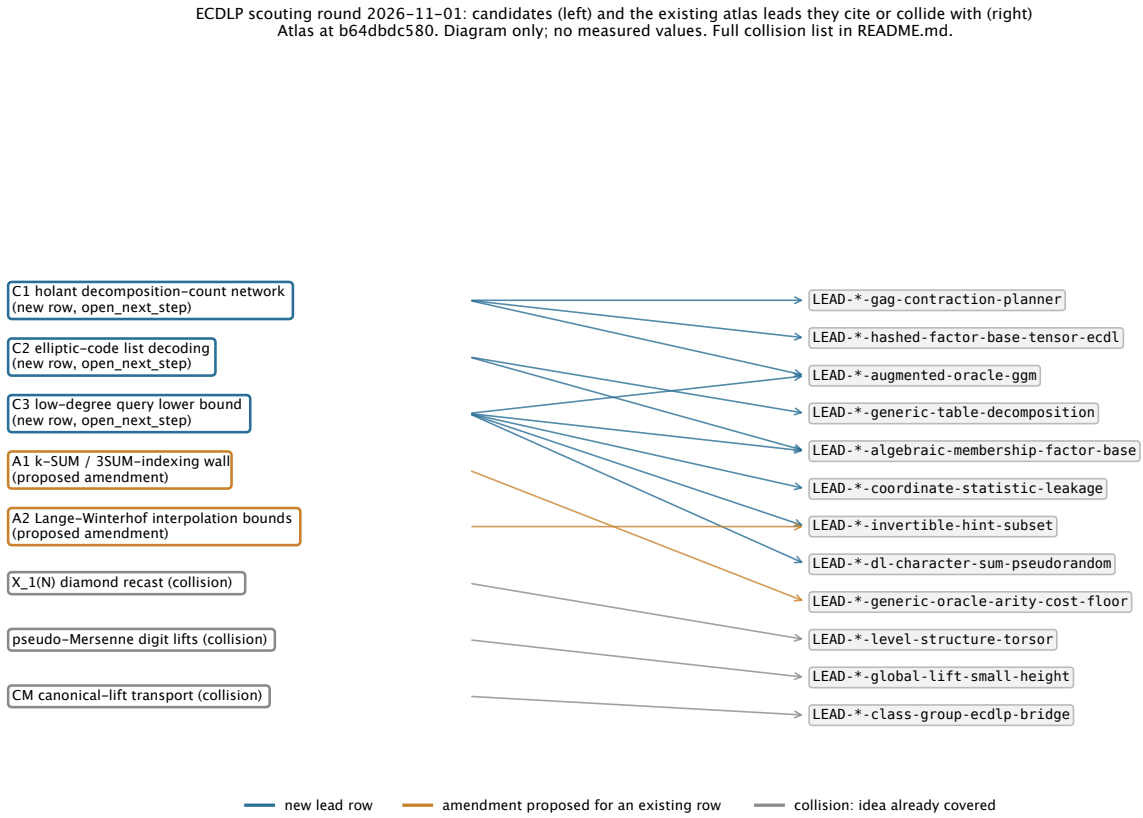



Figure 1: Candidates from this round and the atlas leads they cite or collide with. Atlas at b64dbdc580. Diagram only, no measured values.

## Method

- Read the atlas at `b64dbdc580` (`origin/main`, 2026-11-01), the known-results map, and the standing obstruction summary used in the program's lead-discovery prompt.
- Searched outside fields: diagrammatic algebra and counting complexity, coding theory, fine-grained complexity, additive combinatorics, exponential sums, quantum-circuit simulation.
- Ran novelty gates N1 (`tools/lead_atlas.py --match`) and N2 (`tools/build_frontier_map.py --match`) with at least three phrasings per candidate. Raw output is in `novelty-gates.txt`. The match is lexical, so a miss proves nothing; the top hits were read for every run.
- Gate N3 (ledger index) was not run in this session. The local ledger copy used for validation is behind `main`, so the rows cite no ledger records.

## New lead rows

### C1. Holant view of the decomposition-count network (`LEAD-EXO-holant-decomposition-count-network`)

The count of m-term decompositions of a target R is a closed tensor network over the alphabet F_p. Each node is the ternary tensor `T[x,y,z] = [S_3(x,y,z) = 0]`, and each leaf carries the indicator of the factor base FB.

Contracting one leaf gives the transfer matrix

```
M[y, y'] = #{ x in FB : S_3(x, y, y') = 0 }
```

Up to multiplicity conventions, this is the quotient by negation of the Cayley graph of E(F_p) with connection set {+-F : x(F) in FB}. Its eigenfunctions are the symmetrised characters chi_j + chi_(-j), indexed by the discrete logarithm. A basis that diagonalises M therefore already encodes the answer. The open question is whether any cheaper basis lowers the tensor rank of T.

- **H1.** Entry-access contraction is simulable in the generic group model at O(1) group operations per entry, like the x-oracle in `LEAD-EXO-augmented-oracle-ggm`. Algebraic manipulation of S_3 is not covered.
- **H2.** The Cai-Chen complex-weighted #CSP dichotomy is stated for a fixed finite domain. Here the domain grows with p, so it does not close the lead.
- **Not a duplicate of** `LEAD-ICE-gag-contraction-planner`, which asks which presentation is cheapest for a Groebner solver, or of `LEAD-EXO-hashed-factor-base-tensor-ecdl`, which contracts a Boolean encoding by #SAT.
- **First step.** Prove H1. At p <= 2^10, check the spectrum of M and compare flattening ranks of T under bond-wise basis changes against a random-support null.

### C2. Elliptic-code decoding as a relation finder (`LEAD-ICP-elliptic-code-list-decoding`)

A minimum-weight codeword of the one-point code C_L(FB, kO) vanishes at exactly k factor-base points, and exists exactly when those points sum to O. This is the subset-sum view behind Cheng's NP-hardness result for elliptic code minimum distance. Zhang and Liu (KN-LIT-643) propose finding such codewords by list decoding. The program holds the paper but had no row for it.

- **H1.** A minimum-weight codeword agrees with the zero word in exactly k coordinates. Guruswami-Sudan on a genus-1 code of length B and dimension k guarantees only agreement above about sqrt(B k), which exceeds k whenever B > k. Any received-word strategy must supply the missing agreement, which predicts enumeration-level cost.
- **First step.** Read the full Zhang-Liu text, then run it against k-subset enumeration at p near 2^16 with |FB| near 2^8.

### C3. Low-degree query lower bound (`LEAD-EXO-low-degree-query-lower-bound`)

The generic group model closes named oracles one by one. Polynomial tests on coordinates are oracles it cannot simulate. This lead proposes one bound for the low-degree part of that class, with no auxiliary variables.

- **Scope.** Small-x membership `x(P) < B` has degree B in this model, so it enters only at D about B, where the bound is weaker than sqrt(N). The Legendre symbol of x has degree (p-1)/2. With auxiliary variables, small-x membership drops to degree 2 with about log B extra unknowns, and a bound would have to depend on their number. So this lead does not address index calculus on small-x bases. It targets genuinely low-degree statistics: low-degree curves through tuples of points, cross-ratios, and bounded-degree relations among coordinates of related points.
- **Single-point case.** This is the counting behind `LEAD-ICP-algebraic-membership-factor-base`: a degree-d locus holds O(d) points. That lead was the top N1 hit for this candidate.
- **H1 (conjecture).** With T queries, success probability is at most `O(T^2/N) + O(T D p^(-1/2) polylog p)`, so T is `Omega(min(sqrt N, sqrt p / D))` for D = p^(o(1)).
- **Risk.** The single-point case follows from Weil-type point counts. Adaptive multi-point tuples need character sums along multiples of a point. The Kohel-Shparlinski reference for these is recalled, not verified.
- **First step.** Write the single-test lemma, then attempt the multi-point reduction. In parallel, measure the maximum bias of degree-<=D tests on pairs (P, kP) for p from 2^16 to 2^24 against D / sqrt(p).

## Proposed amendments to existing rows (not edited here)

- **A1, `LEAD-ICE-generic-oracle-arity-cost-floor`.** Its scope excludes oracles that read coordinates algebraically. A conditional fine-grained wall, from 3SUM-indexing or k-SUM with preprocessing, could extend it to RAM-model oracles. Two cautions. The k-SUM conjecture concerns explicit integer lists, while here the set is implicit and the operation is the group law. Kane, Lovett and Moran show k-SUM has near-linear linear-decision-tree complexity, so no query-model proof exists. Suggested step: a literature check of 3SUM-indexing lower bounds (Golovnev et al., STOC 2020, recalled, verify) for implicitly given sets.
- **A2, `LEAD-EXO-invertible-hint-subset`.** Lange and Winterhof (COCOON 2002, LNCS 2387, pp. 137-143; and a 2003 follow-up) prove degree lower bounds for polynomials interpolating the ECDL and ECDH maps. The exact bound was not read in this session. Suggested step: read it before measuring A(d). If it covers agreement on subsets of size at least n^alpha, it decides the algebraic model of that lead from the literature.

## Quick kills, recorded without rows

- **Stabilizer-rank classical simulation of Shor's ECDLP circuit.** Bravyi-Gosset sampling costs about 2^(0.228 t) for t T-gates. Published 256-bit circuits need T or Toffoli counts in the hundreds of millions or more, so the cost is far beyond rho. Related: `LEAD-EXO-quantum-ecdlp-resources`.

- **Shared summation polynomials across a quadratic twist.** A relation on E maps to its twist by scaling x. This helps only when logs on both curves are wanted, so it is Tier I at most.
- **Comparison or ordering oracle on x.** Dropped. Ordering by x is interval membership, which the existing small-x factor bases already use, so it is not a new oracle.

## Collisions with existing leads

| candidate | existing lead |
|---|---|
| GAG presentation planning | `LEAD-ICE-gag-contraction-planner` |
| X_1(N) diamond-operator recast | `LEAD-STR-level-structure-torsor` |
| pseudo-Mersenne digit lifts to a number ring | `LEAD-TRF-global-lift-small-height` |
| CM transport through the canonical lift | `LEAD-STR-class-group-ecdlp-bridge`, `LEAD-TRF-cm-canonical-lift-heights` |
| additive energy of window bases | `LEAD-ICP-coincidence-harvest-structured-base` |
| x-only isogeny-hopping walks | `LEAD-STR-isogeny-class-generic-invariance` |
| split-Jacobian projection of factor bases | `LEAD-EXO-split-jacobian-projection` |
| Eisenstein lattice for the secp256k1 GLV box | `LEAD-STR-endo-scalar-action-generic` |
| spectral / Jennrich decomposition of group tables | `LEAD-RHO-generic-sqrt-lower-bound` |

## Limitations and gaps

- No IDEA records were filed. The propose-ideas flow, ledger-index dedup (N3) and object-frame check (N4) remain for the Coordinator.
- Cheng, Cai-Chen, Lange-Winterhof, Kohel-Shparlinski and the 3SUM-indexing papers have no KN-LIT records yet. Their statements here are recalled or read from abstracts and must be verified before any row relies on them.
- The Conductor coordination server timed out through its MCP tool. The CLI check, run outside the sandbox, returned “Clear”.

## Sources


- Gao, Shaikh, Kissinger, Graphical Algebraic Geometry, arXiv 2605.13993 (KN-LIT-c9e59d).
- Q. Cheng, Hard Problems of Algebraic Geometry Codes, arXiv cs/0507026.
- F. Zhang, S. Liu, Solving ECDLP via List Decoding, ePrint 2018/795 (KN-LIT-643).
- J.-Y. Cai, X. Chen, Complexity of Counting CSP with Complex Weights, arXiv 1111.2384.
- T. Chan, M. Lewenstein, Clustered Integer 3SUM via Additive Combinatorics, arXiv 1502.05204.
- T. Lange, A. Winterhof, Polynomial interpolation of the elliptic curve and XTR discrete logarithm, COCOON 2002.
- S. Bravyi, D. Gosset, Improved classical simulation of quantum circuits dominated by Clifford gates, PRL 116 (2016).


## Files

- `README.md`: this note.
- `novelty-gates.txt`: raw N1 and N2 gate output.
- `figures.py`, `scouting-map.svg`: the candidate-to-lead map.
- `report.typ`, `report.pdf`: rendered report.